

Hardware Business Management System

Purpose of This Document

Defines system components, architecture, technical design principles, and component responsibilities.

Known Project Scope

- Mobile application and web administration system.
- Super Admin and Shop Admin roles.
- Products, purchases, sales, inventory and stock movement.
- Expenses, capital, debts, profit reporting, and 60/40 allocation support.
- Daily opening and daily closing workflows.
- Offline-first mobile storage using SQLite and central MySQL synchronization.
- Notifications, reports, audit trail, validation, error handling, security, and acceptance criteria.

Status Label Convention

Confirmed means the item is part of the agreed scope. Proposed means it is a recommended implementation detail that can still change. Pending Decision means the business owner must decide before a developer or Cloud AI treats it as final.

Status	Meaning	How to Treat It
Confirmed	Agreed project requirement or design direction.	Use as baseline for planning and implementation.
Proposed	Recommended approach that may still change.	Validate during design and owner review.
Pending Decision	Business, accounting, or technical rule not fully approved.	Do not hard code. Ask the owner or make configurable.

Architecture Summary

Component	Responsibility	Status
Mobile app	Daily shop workflows, offline SQLite storage, sync queue, notifications, closing.	Confirmed

Component	Responsibility	Status
Web admin	Central management, reporting, configuration, audit review, exports.	Confirmed
Backend API	Authentication, authorization, validation, business rules, sync, reports.	Proposed
SQLite	Local mobile persistence for offline-first behavior.	Confirmed
MySQL	Central data source for synchronized records and reports.	Confirmed
Sync service	Transfers local pending records and returns server acknowledgements.	Confirmed

Technical Design Principles

- Separate mobile, web, backend, database, sync, reporting, and notification responsibilities.
- Use role and shop permission checks in both UI and backend.
- Use append-only transaction and movement histories where business traceability matters.
- Use stable local IDs for offline records so retry operations are safe.
- Keep unresolved accounting rules configurable or blocked behind owner approval.

Open Decisions That Must Not Be Invented

Decision Area	What Must Be Confirmed	Status
Profit formula	Exact formula for gross profit, net profit, cash profit, and expense treatment.	Pending Decision
Stock costing method	Latest cost, weighted average, FIFO, or another method for cost of goods sold.	Pending Decision
60/40 destination labels	Meaning of 60 percent and 40 percent amounts in business reports.	Pending Decision
Allocation timing	Daily, weekly, monthly, per closing, or owner-approved timing.	Pending Decision
Tax or VAT handling	Whether tax calculation is part of first release and how it affects reports.	Pending Decision
Returns refunds discounts	How returns, damaged goods, refunds, and discounts affect stock and profit.	Pending Decision

Acceptance of This Document

- The document is acceptable when the owner can identify which items are confirmed, proposed, and pending decision.
- A developer or Cloud AI can use the confirmed parts without needing to guess the project direction.
- Pending accounting rules remain visible and are not presented as final business rules.